Predicting Dog Adoptions Using a Machine Learning Model

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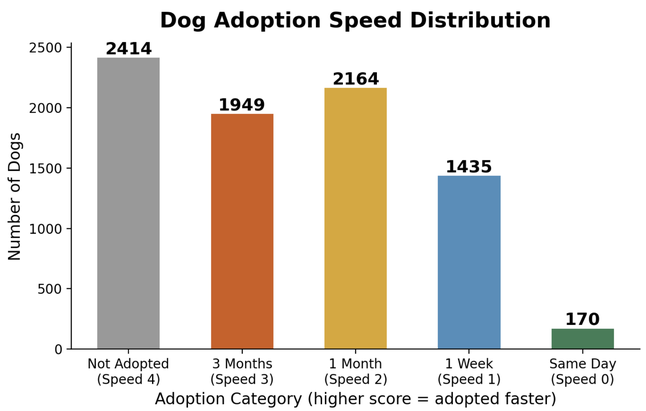
Objective

This project predicts how quickly a dog listed on PetFinder will be adopted. The model outputs a continuous adoption score between 0 and 1 that is easier to interpret than the raw category.

Score near 1.0 = likely adopted same-day or within a week
Score near 0.0 = unlikely to be adopted within 100 days

The two main techniques applied are gradient boosted trees (LightGBM) and natural language processing (VADER sentiment analysis).



Data & Features

The dataset comes from the PetFinder Adoption Prediction competition on Kaggle. The training set contains 8,132 dog listings after filtering to dogs only.

18 features used including Age, Breed, Gender, Color, MaturitySize, FurLength, vaccination status, photo count, adoption fee, and 4 VADER sentiment scores extracted from rescuer descriptions.

Health and VideoAmt were dropped after showing near-zero importance.

Model & Methods

LightGBM Regressor

A gradient boosting framework using histogram-based learning and leaf-wise tree growth. Trained with objective=regression and metric=RMSE. Early stopping with patience of 50 rounds, tuned via 3-fold cross-validation. Final model: 634 trees.

Feature Importance (Top 7)

Feature Importance	Gain
Age	2305
sentiment_compound	1394
Breed1	1271
sentiment_neu	1208
sentiment_pos	1135
PhotoAmt	856
sentiment_neg	782

NLP & Sentiment Analysis

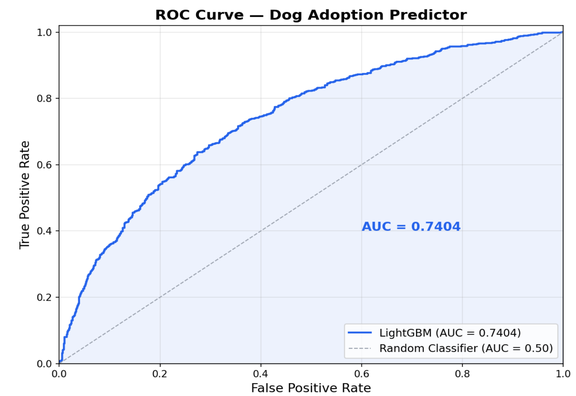
VADER (Valence Aware Dictionary and Sentiment Reasoner) is a rule-based tool that scores text against a pre-built lexicon accounting for punctuation, capitalization, and modifiers.

Each description produced 4 scores: negativity, neutrality, positivity, and compound (-1 to +1). All four ranked in the top 7 features by information gain, with compound ranking #2 behind Age.

Dogs with more positive, expressive descriptions tended to get adopted faster.

Results

The ROC AUC of 0.74 means the model correctly ranks a fast-adoption dog above a slow-adoption dog about 74% of the time. The modest R^2 reflects that adoption speed depends on factors the data cannot capture.



Key Findings

Age is the strongest predictor by a wide margin. Younger dogs get adopted faster.

Sentiment scores from VADER outperform most tabular features. Dogs with more positive descriptions get adopted faster.

Photo count ranked in the top 6. More photos correlate with faster adoption.

References

[1] PetFinder Adoption Prediction, Kaggle Competition. https://www.kaggle.com/competitions/petfinder-adoption-prediction

[2] G. Ke et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," NeurIPS 2017.

[3] C. J. Hutto and E. Gilbert, "VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text," ICWSM 2014.

[4] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," JMLR 2011.

github.com/JaydenCruz2004/Dog-Adoption-Predictor